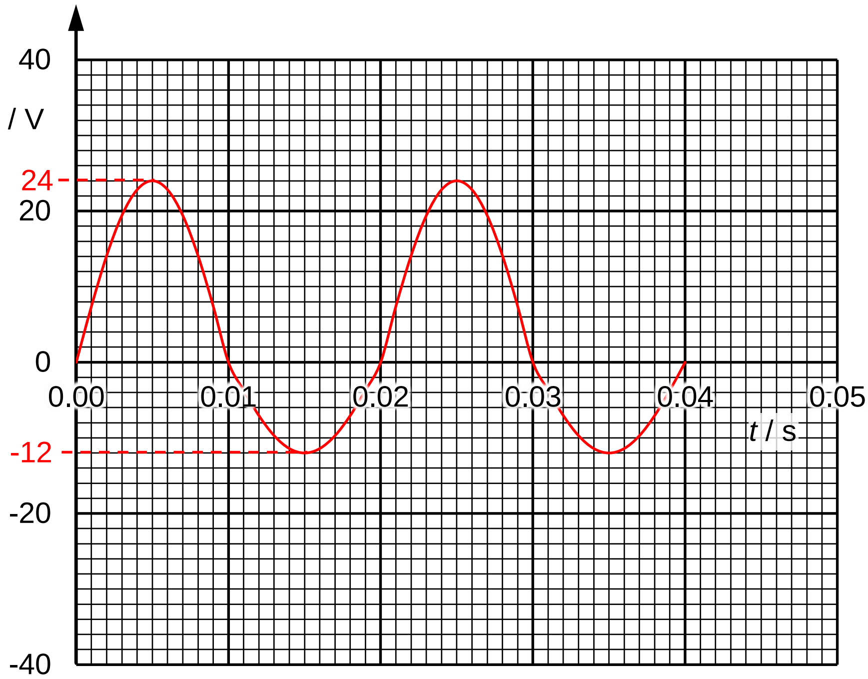
6 **(a)** The root-mean-square value of an alternating current is that <u>value of the direct current that would produce thermal energy at the same rate in the same resistor</u>.

(b) **(i)**



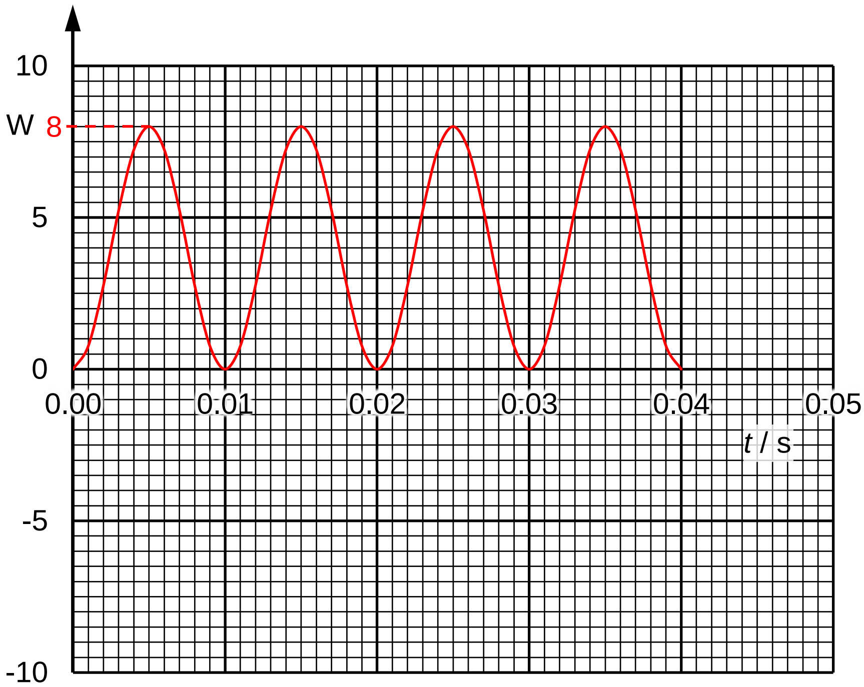
With S closed, current passes through the diode and R_1 in the forward bias direction. In the reverse bias direction, current passes through R_1 and R_2.

Forward bias: peak V_1 = peak $V = 24$ V
Reverse bias: peak $V_1 = 12$ V

period $T = \dfrac{2\pi}{\omega} = \dfrac{2\pi}{314} = 0.02$ s

*Sine graph with correct period.

(ii)



With S opened, current passes through R_1 and R_2 all the time and does not pass through the diode.
Since V and I of the alternating supply is sinusoidal, power graph will be a sine^2 graph.

peak $P_1 = \dfrac{(\text{peak } V_1)^2}{R_1} = \dfrac{(24/2)^2}{18} = 8\text{ W}$

*Correct shape (smooth curve, sine squared) with correct period.

7 **(c)** The emission line spectrum consists of a series of distinct lines of specific colours/